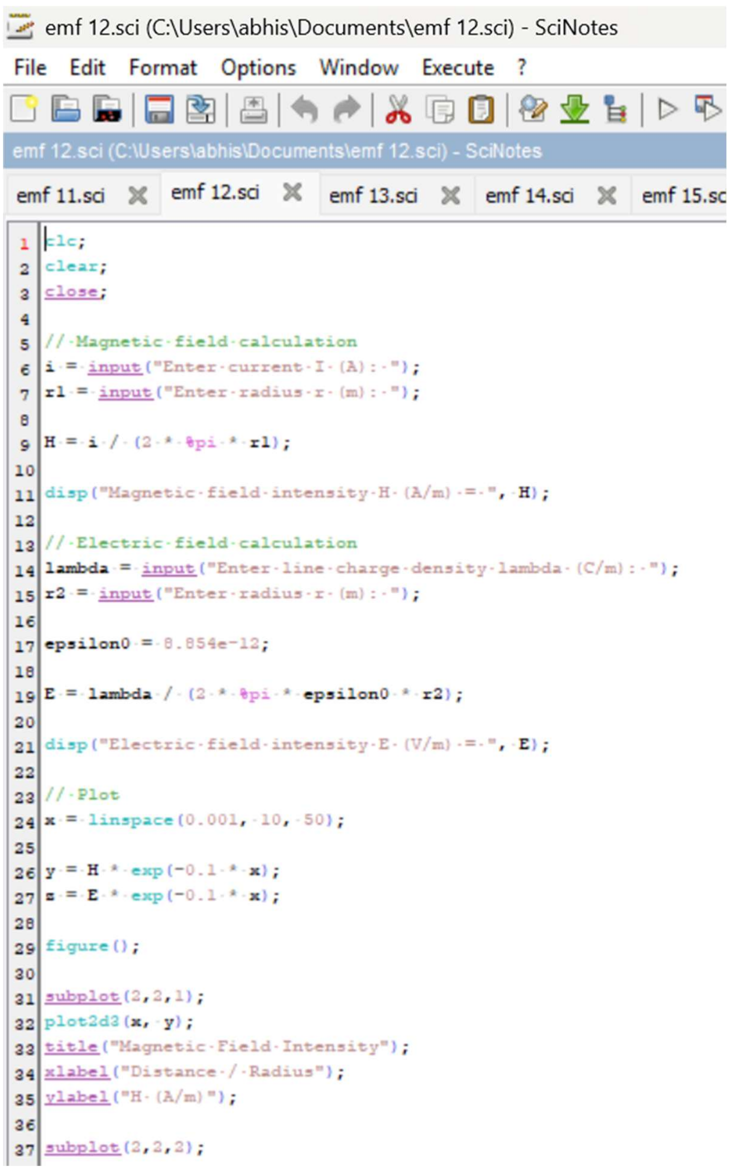
Exp No:12 Determination and plotting of magnetic and electric field intensity due to infinite long conductor at any specific point located at H

Aim: To write program for Magnetic field intensity of infinite long conductor using SCILAB.

Software required:

• SCILAB version 6.0.1

Theory: In electromagnetism, the fields around a conductor carrying a current are fundamental concepts. For an infinite long conductor carrying a current, we can determine both the magnetic field intensity and electric field intensity at a point located at a distance from the conductor.




```
clc;
clear;
close;

// Magnetic field calculation
i = input("Enter current I (A): ");
r1 = input("Enter radius r (m): ");

H = i / (2 * %pi * r1);

disp("Magnetic field intensity H (A/m) = ", H);

// Electric field calculation
lambda = input("Enter line charge density lambda (C/m): ");
r2 = input("Enter radius r (m): ");

epsilon0 = 8.854e-12;

E = lambda / (2 * %pi * epsilon0 * r2);

disp("Electric field intensity E (V/m) = ", E);

// Plot
x = linspace(0.001, 10, 50);

y = H * exp(-0.1 * x);
z = E * exp(-0.1 * x);

figure();

subplot(2,2,1);
plot2d3(x, y);
title("Magnetic Field Intensity");
xlabel("Distance / Radius");
ylabel("H (A/m)");

subplot(2,2,2);
```

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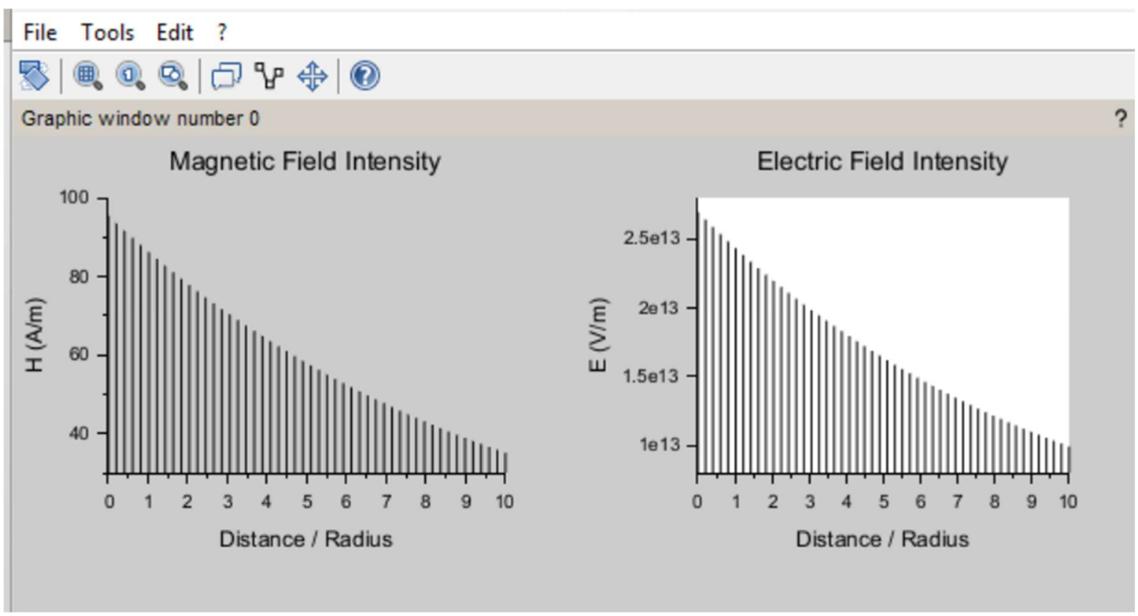
```
Enter current I (A): 3

Enter radius r (m): 5*10^-3

  "Magnetic field intensity H (A/m) = "
   95.492966
Enter line charge density lambda (C/m): 6

Enter radius r (m): 4*10^-3

  "Electric field intensity E (V/m) = "
   2.696D+13
exec: Wrong number of output argument(s): 0 expected.

-->
```



$i = 3$

$r = 5\,mm$ $\qquad d = 6m$

$r = 6\,mm$

$$H = \frac{I}{2\pi r}$$

$$H = \frac{3}{2(3.14159)(5\times10^{-3})}$$

$$\boxed{H = 9.5.493\ A/m}$$

result: $95.492966\ A/m$

o/p

$d = 6\,dm$ $\quad r = 4\times10^{-3}m$

$\varepsilon_0 = 8.854\times10^{-12}F/m$

$$E = \frac{d}{2\pi\varepsilon_0 r}$$

$$E = \frac{6}{2(3.14159)(8854\times10^{-12})(4\times10^{-3})}$$

$$\boxed{E = 2696\times10^{3}\ V/m}$$

$2.696D+13\ V/m$ //.

Case 1: To examine how the magnetic and electric field intensities change with varying distances from an infinitely long current-carrying conductor.

12.01.sci (C:\Users\abhis\Documents\12.01.sci) - SciNotes

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```
clc;
clear;

// Constants
mu_0 = 4 * %pi * 10^(-7);       // Permeability of free space (T*m/A)
epsilon_0 = 8.854 * 10^(-12);   // Permittivity of free space (F/m)
I = 10;                         // Current in Amperes

// Distances from the conductor
distances = [0.1, 1, 5, 10];    // in meters
num_distances = length(distances);

// Initialize arrays
B_field = zeros(1, num_distances);
E_field = zeros(1, num_distances);

// Calculate magnetic and electric fields
for i = 1:num_distances
    r = distances(i);

    // Magnetic field
    B_field(i) = (mu_0 * I) / (2 * %pi * r);

    // Electric field
    E_field(i) = I / (2 * %pi * epsilon_0 * r);
end

// Display results
disp("Distance (m) | Magnetic Field (T) | Electric Field (N/C)");

for i = 1:num_distances
    mprintf("%g | %g | %e\n", distances(i), B_field(i), E_field(i));
end
```

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```
  "Distance (m) | Magnetic Field (T) | Electric Field (N/C)"
0.1 | 2e-05 | 1.797548e+12
1 | 2e-06 | 1.797548e+11
5 | 4e-07 | 3.595097e+10
10 | 2e-07 | 1.797548e+10

-->
```

Case 2:To study how the magnetic and electric field intensities change for an infinitely long conductor ata fixed distance H but with varying currents.

12.3.sci (C:\Users\abhis\Documents\12.3.sci) - SciNotes

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12.3.sci (C:\Users\abhis\Documents\12.3.sci) - SciNotes

emf 11.sci | emf 12.sci | emf 13.sci | emf 14.sci | emf 15.sci | 15.sci | 11.2.sci

```
clc;
clear;

// Constants
H = 2;                    // Distance from the conductor in meters
epsilon_0 = 8.854e-12;    // Permittivity of free space (F/m)
lambda = 1.0e-6;          // Linear charge density (C/m)

// Currents to evaluate
currents = [5, 10, 20, 30, 50];
num_currents = length(currents);

// Preallocate arrays
magnetic_field_intensity = zeros(1, num_currents);
electric_field_intensity = zeros(1, num_currents);

// Calculate magnetic and electric field intensities
for i = 1:num_currents
    
    I = currents(i);
    
    // Magnetic field intensity
    magnetic_field_intensity(i) = I / (2 * %pi * H);
    
    // Electric field intensity
    electric_field_intensity(i) = lambda / (2 * %pi * epsilon_0 * H);
    
end

// Display results
disp("Current (A)    Magnetic Field Intensity (A/m)    Electric Field Intensity (V/m)");

for i = 1:num_currents
    mprintf("%10.2f        %20.6f        %25.6e\n", ...
            currents(i), ...
            magnetic_field_intensity(i), ...
            electric_field_intensity(i));
```

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```
 "Current (A)    Magnetic Field Intensity (A/m)    Electric Field Intensity (V/m)"
      5.00                0.397887                 8.987742e+03
     10.00                0.795775                 8.987742e+03
     20.00                1.591549                 8.987742e+03
     30.00                2.387324                 8.987742e+03
     50.00                3.978874                 8.987742e+03

-->
```

Case 3:To analyze the magnetic and electric field intensities due to an infinitely long conductor when there is a potential difference (voltage) across the conductor, simulating a capacitive effect.

12.03.sci (C:\Users\abhis\Documents\12.03.sci) - SciNotes

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12.03.sci (C:\Users\abhis\Documents\12.03.sci) - SciNotes

emf 11.sci emf 12.sci emf 13.sci emf 14.sci emf 15.sci 15.sci

```
mu_0 = 4 * %pi * 1e-7;      // Permeability of free space (H/m)
R = 10;                     // Resistance in ohms

// Voltage values
V_values = [100, 300, 500, 700, 1000];  // Voltage in volts
H_distance = 1;                         // Distance from conductor (m)

// Initialize arrays
E_values = zeros(1, length(V_values));
H_values = zeros(1, length(V_values));

// Calculate Electric Field (E) and Magnetic Field Intensity (H)
for i = 1:length(V_values)
    
    V = V_values(i);
    
    // Current using Ohm's Law
    I = V / R;
    
    // Assume linear charge density is proportional to current
    lambda = I;
    
    // Electric field
    E_values(i) = lambda / (2 * %pi * epsilon_0 * H_distance);
    
    // Magnetic field intensity
    H_values(i) = I / (2 * %pi * H_distance);
    
end

// Display results
disp("Voltage (V)    Electric Field (E)    Magnetic Field (H)");

for i = 1:length(V_values)
    mprintf("%10.2f    %20.6e    %20.6f\n", ...
            V_values(i), E_values(i), H_values(i));
end
```

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```
 "Voltage (V)    Electric Field (E)    Magnetic Field (H)"
    100.00          1.797548e+11              1.591549
    300.00          5.392645e+11              4.774648
    500.00          8.987742e+11              7.957747
    700.00          1.258284e+12             11.140846
   1000.00          1.797548e+12             15.915494

-->
```

Result: Thus the program was verified for the magnetic and electric field intensities due to infinitely long conductor.